Conditional Image-Level Closure in the Q-RAIF Framework

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Abstract

This addendum isolates an algebraic interface between two modelling roles in the **Q-RAIF (Quantum Reference Algebra for Information Flow)** framework. Paper A [1] adopts a $Cl(1,3)$ representation as a Lorentzian-encoding choice under supplied quadratic and generator data. Paper B [2] motivates a Cliffordian parameterization $Cl(V,q)$ for a control algebra under its stated compatibility conditions; neither paper uniquely derives Clifford structure.

Given a supplied unital $*$-homomorphism from the control algebra to the environmental algebra, we establish an **image-level closure result**: its image is a subalgebra, and in the stated even-dimensional Clifford case a nonzero homomorphism is injective. The *Same-Clock* statement records an operational co-indexing condition; it is not an input to the image proof and does not by itself ensure thermodynamic potency.

This note does not derive the representation map, select a preferred $Cl(1,3)$ subalgebra, or prove signature preservation. Physical realizability therefore remains an open interface rather than a completed bridge.

Keywords: Q-RAIF, realizability, representation, operator algebra, Clifford algebra, open quantum systems, Lyapunov stability, algebraic closure

1 Introduction

1.1 Context: The Q-RAIF Program

The Quantum Reference Algebra for Information Flow (Q-RAIF) framework investigates algebraic representations for geometric read-out and subsystem control. Revised Papers A and B treat Clifford structure as a modelling input rather than a necessary consequence. The programme builds on the conditional algebra/reconstruction setting of HAFF [3, 4].

Paper	Question	Analogy	Result
HAFF [3]	How does geometry emerge?	Ocean	Algebra $\to$ Geometry
Q-RAIF A [1]	How encode Lorentzian read-out?	Water	$Cl(1,3)$ model
Q-RAIF B [2]	How parameterize controls?	Fish	$Cl(V,q)$ model
This work	What follows from representation?	Bridge	$\mathrm{Im}\,\phi \subseteq \mathcal{A}_{\mathrm{ext}}$

1.2 The Logical Gap

Papers A and B both posit Clifford representations for different tasks; they do not independently derive them [1, 2]. There is therefore no convergence theorem to close. This note isolates the tautological algebraic consequence of an explicit representation input and identifies the still-open physical question: what dynamics, if any, supplies that input?

1.3 Scope

This addendum explicitly assumes a representation map from an internal model algebra into an external observable algebra. It does not derive that map from persistence or system–environment coupling. Its image statement is algebraic; physical realizability remains an interface problem.

Result-status ledger. The representation map is an interface assumption. Same-clock overlap is an operational remark and does not enter the image proof. Semisimplicity/injectivity is a standard finite-dimensional Clifford-algebra fact. The former closure theorem is downgraded to an image-level proposition; all signature compatibility claims require additional quadratic-form-preserving data.

2 Setup and Prerequisites

Let $\mathcal{U}$ be a universe described by the Q-RAIF framework.

- **Environment ("water").** Let $\mathcal{A}_{\mathrm{ext}}$ denote the algebra of observables accessible at the holographic boundary. Paper A [1] uses a $Cl(1,3)$ representation as a model for a supplied Lorentzian read-out; it does not prove that every $\mathcal{A}_{\mathrm{ext}}$ must contain a preferred copy.
- **Subsystem ("fish").** Let $\mathcal{O}_{\mathrm{int}}$ denote the internal control algebra of a persistent subsystem $R \subset \mathcal{U}$. Paper B [2] may parameterize controls by $\mathcal{O}_{\mathrm{int}} \cong Cl(V,q)$, but persistence does not force that choice.

The remaining logical gap is the relationship between $\mathcal{O}_{\mathrm{int}}$ and $\mathcal{A}_{\mathrm{ext}}$: can a stable Clifford control algebra exist while being structurally disjoint from the available environmental observables?

3 Realizability and Same-Clock Co-Indexing

Definition 1 (Algebraic Representation Input)**.** *A control algebra $\mathcal{O}_{\text{int}}$ is represented in an environment $\mathcal{A}_{\text{ext}}$ when a unital algebra homomorphism is supplied,*

$$\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}} \tag{1}$$

such that $\text{Im}(\phi)$ *has non-zero action on the interaction Hamiltonian H_{int}, i.e.,* $[\text{Im}(\phi), H_{\text{int}}] \neq 0$*. The commutator checks nontrivial coupling once ϕ is given. The definition does not prove that persistence or coupling produces such a homomorphism; calling it physical realizability is therefore an interface assumption.*

Remark 2 (Same-clock operational condition)**.** *Let I be an operational index set and suppose a model supplies a net $J \mapsto \mathcal{A}(J)$ for operational windows $J \subseteq I$. A feedback protocol may require windows $J_{\text{ctrl}}, J_{\text{env}} \subseteq I$ with*

1. ***Non-null intersection:*** *$J_{\text{ctrl}} \cap J_{\text{env}} \neq \emptyset$.*
2. ***Window integrity:*** *every critical lookback window $W \subseteq J_{\text{ctrl}} \cap J_{\text{env}}$ has a declared algebra $\mathcal{A}(W)$ and feedback channel.*

These are reasonable protocol requirements for discussing entropy export [8], but they neither imply the existence of ϕ nor enter the algebraic image proposition below. The former Same-Clock lemma is therefore demoted to this operational remark.

3.1 Semisimplicity of Clifford Algebras

Lemma 3 (Injectivity from Semisimplicity)**.** *Let (V, q) be a finite-dimensional real vector space with non-degenerate quadratic form. Then $Cl(V, q)$ is semisimple. If* $\dim V$ *is even, $Cl(V, q)$ is simple, and every non-zero algebra homomorphism $\phi : Cl(V, q) \to \mathcal{A}$ is injective.*

Proof. By the periodicity theorem for real Clifford algebras [5, 7], $Cl(V, q)$ with non-degenerate q is isomorphic to a matrix algebra $M_n(K)$ or a direct sum $M_n(K) \oplus M_n(K)$, where $K \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ depends on the signature and dimension modulo 8. In either case the algebra is semisimple.

When $\dim V$ is even, $Cl(V, q)$ is simple (a single matrix block). The kernel of any algebra homomorphism is a two-sided ideal; a simple algebra admits no proper non-trivial ideals, so $\ker \phi = \{0\}$ whenever $\phi \neq 0$. □

4 The Image Proposition

Proposition 4 (Image of a represented Clifford algebra)**.** *Let $\mathcal{O}_{\text{int}} \cong Cl(V, q)$ be a chosen finite-dimensional Clifford model and let $\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}}$ be the homomorphism supplied by Definition 1. Then*

$$\mathcal{O}_{\text{eff}} := \text{Im}(\phi) \subseteq \mathcal{A}_{\text{ext}} \tag{2}$$

is a subalgebra satisfying the induced quotient relations. If $\dim V$ *is even and $\phi \neq 0$, then ϕ is injective and $\mathcal{O}_{\text{eff}} \cong Cl(V, q)$.*

Proof. The image of an algebra homomorphism is a subalgebra, and relations in the source descend to its image. Lemma 3 gives injectivity for even $\dim V$ when the homomorphism is nonzero. No Same-Clock hypothesis is used. □

Remark 5 (Scope of the image proposition)**.** *The logical structure of Proposition 4 is simply: "the image of a homomorphism is a subalgebra," plus the even-dimensional injectivity fact. Papers A and B no longer supply independent selection theorems. The physically nontrivial step—deriving a representation ϕ from system–environment dynamics—is assumed rather than proved. Operator-algebra inclusion, virtual-subsystem, split-property, or modular-inclusion techniques may inform a future derivation [9, 10, 11].*

Withdrawal of the "No Ghost Algebra" corollary. The statement that a non-represented algebra fails Definition 1 merely repeats the definition. Its former signature obstruction is false for ordinary algebra homomorphisms: matrix-algebra embeddings need not preserve a presentation's quadratic form or signature.

To compare signatures one must supply a quadratic-space morphism $j : (V, q) \to (W, Q)$ satisfying

$$Q(jv) = q(v) \tag{3}$$

and, where relevant, require the induced Clifford map to preserve the unit, involution, and grading. Proposition 4 assumes no ambient quadratic space (W, Q) and proves no such isometry. It therefore yields an image in $\mathcal{A}_{\text{ext}}$, not an embedding into a distinguished $Cl(1, 3)$ subalgebra and not a signature-compatibility theorem.

5 Discussion

5.1 What This Result Does and Does Not Show

Does show: Given a supplied homomorphism, its image is a subalgebra of $\mathcal{A}_{\text{ext}}$; for even-dimensional $Cl(V, q)$, a nonzero homomorphism is injective.

Does not show: That ϕ must be injective in general—however, by Lemma 3, injectivity *is* guaranteed when $\dim V$ is even (which includes the physically relevant case $\dim V = 4$). For odd $\dim V$, the image $\mathrm{Im}(\phi)$ is isomorphic only to the quotient $Cl(V, q)/\ker\phi$; depending on the real signature and the representation, this quotient may be the full simple algebra or one surviving simple block of a semisimple direct sum. No general injectivity or preservation of the full Clifford presentation follows in the odd-dimensional case. That the image lies in a preferred $Cl(1, 3)$ subalgebra or preserves signature. That persistence produces ϕ or selects Clifford structure.

5.2 The Bridge Statement

Remark 6 (Closing the Loop)**.** *Papers A and B propose Clifford representations for separate modelling roles. Proposition 4 does not lock them together independently; it records what follows after a representation map is supplied. Deriving that map and any quadratic compatibility is the open bridge problem.*

5.3 Connection to HAFF

Within the HAFF programme [3, 4], geometric reconstruction is conditional on selected observable-algebra and dictionary data. The image proposition adds no independent natural-selection result. It isolates the representation/interface assumption that a future dynamical construction would have to justify.

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